

Mate 6551

Tarea 2

- 1.9. If $f: X \rightarrow Y$ is nullhomotopic and if $g: Y \rightarrow Z$ is continuous, then $g \circ f$ is nullhomotopic.
- 1.14. Prove that $X \mapsto CX$ defines a functor $\mathbf{Top} \rightarrow \mathbf{Top}$ (the reader must define the behavior on morphisms). (*Hint*: Use Exercise 1.11.)
- *1.21. If X and Y are path connected, then $X \times Y$ is path connected.
- *1.22. If $f: X \rightarrow Y$ is continuous and X is path connected, then $f(X)$ is path connected.
- *1.24. The path components of a space X are maximal path connected subspaces; moreover, every path connected subset of X is contained in a unique path component of X .